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ProtoExplain-LM: A Prototype-Guided Explainable Vision–Language Framework for Trustworthy Plant Leaf Disease Identification

Abstract-In a word, it is the necessity to identify the plant leaf disease in an accurate way apart from explainable for sustainable agriculture, early diagnosis and circular of information for farmers as well as agronomists. Despite the promising classification performance, the recent deep learning methods have had little practical impact due to their lack of interpretability, sensitivity under real-world imaging conditions and actionable explanations. Furthermore, the available explainable AI (XAI) methods are generally limited to qualitative saliency maps and do not furnish case-based evidence or agronomically meaningful interpretation. To rectify these gaps, we present the method ProtoExplain-LM, a new prototype-assisted explainable vision–language pipeline for plant leaf disease recognition. The proposed approach forms a light- weight convolutional feature extractor that is connected with a class-specific prototype learning, which supports case-based visual reasoning by comparing discriminatory leaf areas to learned disease prototypes. In contrast to typical black-box models, ProtoExplain-LM is explicitly grounded for each prediction on a small number of representative visual prototypes. A large language model (LLM) is used for structured explanation generation: it maps prototype activations, certainty estimates, and context metadata to brief diagnostic explanations and treatment recommendations in natural language. Extensive experiments on benchmark PlantVillage Dataset indicate that ProtoExplain-LM achieves 98.12% classification accuracy outperforming the traditional CNN based models by 4.6% under cross-dataset evaluation. The proposed framework further achieves an explanation fidelity of 92.4%, signifying high consistency between visual evidence and model prediction, along with low inference latency for edge deployment. These findings demonstrate the reliability and interpretability of ProtoExplain-LM to provide reliable, transparent and actionable plant disease diagnosis, linking state-of-the-art performance with on-farm agronomic trustworthiness.

Keywords- Plant leaf disease identification, Prototype learning, Explainable artificial intelligence, Vision–language models, Large language models, Deep learning, Precision agriculture

I Introduction

Leaf diseases are primary threats to the global agricultural yield due to direct loss in quality and quantity of crops leading to food epidemiological concern [1], [2]. Early disease detection is one of the most important and effective methods for precision agriculture, because it can facilitate rapid intervention and then minimize economic loss [3], [4]. Driven by increasing application of digital farming solutions, image based plant disease detection has been considered as a practical and scalable alternative to conventional field level inspection approaches [18], [21]. However, accurate disease monitoring is still a challenging task as it is affected by illumination changes, complex background, visual similarity of diseases and the limited number of expert agronomists especially in remote and resource-constrained areas [8], [19].

Conventional diagnosing of plant disease depends heavily on visual inspection by farmers or experts [5], [6]. It is a time-consuming and subjective process, with high human errors. Furthermore, manual methods are not adequate for large agriculture areas or a variety of crops. To resolve these challenges, deep learning models, especially CNNs, have been extensively employed for automatic plant leaf disease identification. Recent works demonstrate that CNN-based and transfer learning models can reach high accuracy in several crops and disease categories [1], [3], [7], [10]. More advanced architectures, such as DenseNet [6], EfficientNet [16], and attentionbased models[38], which further enhance robustness as well as generalize in real world scenarios. Despite their good results, the real applicability of these models is still reduced by their black-box nature.

One notable limitation of most prevailing deep learning models is that they are in the form of black-box predictor, which can only provide prediction outputs without showing why a decision is made. The absence of interpretability undermines the user's confidence, which is especially important in agricultural decision-making due to potential erroneous diagnoses that cause false treatments and higher degree of damage on crop [21], [22]. Finally, in [36], National Institute of Food and Agriculture (NIFA) of the United States Department of Agriculture-funded plant disease detection systems have been proposed with XAI methods including Grad-CAM [38] and saliency map visualization for increased transparency. Although such methods emphasize the critical image parts, they are generally qualitative and have little agronomic significance. Moreover, saliency based explanations are not case-based reference or practical methods for farmers to understand and use directly [19][40].

Another drawback of popular methods is the lack of case-based reasoning, which is often applied by agricultural experts in disease diagnosis [15]. Disease diagnosis in practice is usually based on the comparison of observed symptoms with previous cases, which does not fit for most of deep learning models. It has been observed that the plant-disease identification systems currently available, seldom promote natural-language explanations which are easily understandable regarding their diagnosis generation process. Despite the fact that LLMs are powerful in reasoning and explainable generation, their utility for integrating visual disease evidence with agricultural setting has not been fully investigated.

To meet these challenges, we present ProtoExplain-LM, a prototype-lead explainable vision–language technique for accurate plant leaf disease diagnosis. The main contributions of our work are:

(i) a prototype-guided visual reasoning, where disease predictions are connected to prototypical leaf representations for transparency and case-based explanation;

(ii) a vision–language integration system, which uses an LLM to produce clear human-readable diagnostic explanations and agronomic recommendations based on visual evidence.

(iii) comprehensive experimental evaluation that shows the effectiveness of our approach in learning more discriminative features and better explaining image classification together with low inference time for practical and on-edge agricultural applications.

1.1 Objectives of the Research

- To build a prototype-based vision–language model to achieve efficient and interpretable plant leaf disease identification.
- To produce coherent natural-language disease explanations based on visual prototype evidence and a large language model.
- To substantiate the performance of the proposed method in accuracy, explainability, and inference speed.

1.2 Organization of the paper

The rest of this paper is divided as follows. In Section II, we summarise the related work on identification of plant leaf disease and motivate for this study. The architecture of the proposed ProtoExplain-LM is detailed in Section III, experimental setup and result analysis are presented in sections IV and V respectively and section VI concludes the paper with summary and discussion.

II Related Works

Abd Algani et al. (2023) proposed ACO-CNN that adopted an optimization-based deep learning model to predict plant's leaf diseases using Ant Colony Optimization (ACO) and Convolutional Neural Network (CNN). They focused on increasing disease classification accuracy by enhancing feature information from the color, texture and leaf geometry aspects. The proposed pipeline, consisting of image acquisition-segmentation-noise reduction-CNN based classification, was found to outperform existing methods for the diagnosis task and this work underlined that bio-inspired optimization can be effective in plant disease diagnosis [1]. Pandey and Jain (2022) tackled the failure of CNN's robustness in real world imaging conditions by introducing Dense attention dense convolutional neural network (DADCNN-5). Their model augmented the feature discrimination by applying the mixed sigmoid attention into dense learning block, which could let the network emphasize in lesion region and suppress background noise. Extensive experimental results demonstrated the high quality of classification accuracy and robustness, indicating that attention mechanisms are essential to the practical disease identification systems [2]. Fan et al. (2022) proposed a hybrid feature learning approach using learn-based deep features with hand-crafted texture descriptors by feature fusion. With the help of center loss, the approach increased intra-class compactness and inter-class separability, leading to an optimal disease discrimination over multiple crop datasets. They showed that fusion of deep and hand-crafted features enhance generalization for plant disease classification [3]. Gajjar et al. (2022) concentrated on detection and localization of disease in real-time using a COVID-19 (CT/OR) screening system developed using CNN-based classification and localization framework deployed on an embedded NVIDIA Jetson TX1 platform. By incorporating a one-step detector, we can identify and localize the diseases under a field environment with high accuracy, which indicates that deep learning-based models can be applied to real agricultural environments [4].

Pal and Kumar, 2023 was also the first framework that combined Inception–VGG hybrid backbone with Kohonen based deep learning for disease severity classification. They adopted image preprocessing, occlusion treatment patterned on multivariate GrabCut segmentation and transfer learning to reduce the issue of overfitting. Compared with [5], the performance of the proposed system was relatively better in terms of accuracies and robustness for real-time images from the agricultural backgrounds containing complex backgrounds. Nandhini and Ashokkumar (2022) presented mutation-based Henry Gas Solubility Optimization algorithm for tuning the hyperparameters of DenseNet-121. Their optimized ConvNet model decreased computational cost with high- er classification accuracy on both PlantVillage and real-field datasets. The contribution emphasized the use of evolutionary optimization to improve CNN efficiency and stability for plant disease diagnosis [6]. Ashwinkumar et al. (2022) introduced an automated disease diagnostic system with an optimized MobileNet-based CNN model, while utilizing hyperparameters optimization applied the Emperor Penguin Optimizer. The scheme consisted of preprocessing, segmentation, feature extraction and classification stages, and showed a high precision-recall performance. Their results showed that lightweight CNNs could be used for large-scale agricultural monitoring [7]. Harakannanavar et al. (2022) studied the classic and deep learning models using tomato leaf deterioration detection. Through combining image preprocessing, clustering-based segmentation and texture (wavelet) feature, the research compared classification performance of SVM, K-NN and CNN. The performance of CNN was superior to the conventional approach in this study, emphasizing the superiority of deep learning model in plant disease phenotyping [8]. Alqahtani et al. (2023) introduced PlantRefineDet, a better one-stage detection model obtained by establishing the backbone of RefineDet to ResNet-50. Their approach achieved better disease localisation and classification by removing anchor refinement, using the features multiple times. Testing results on the PlantVillage dataset showed that near-perfect recognition accuracy, bringing out the significance of detection-based architectures in recognizing plant disease [9].

Shewale and Daruwala (2023) introduced a CNN based disease classification model working on live tomato leaf images acquired under challenging environmental conditions. Their method avoided hand-crafted feature engineering and directly used complex discriminator based feature representations, obtaining competitive accuracy and stability. The research demonstrated the importance of RWSD to test deep learning based algorithms in agriculture too [10]. Kaur et al. (2023) proposed another modified InceptionResNet-V2 network along with transfer learning to categorize tomato leaves diseases. Their method was able to overcome the shortcoming of mixed-crop datasets, and showed that it could achieve high classification accuracy over various disease classes. The research demonstrated the success of deep transfer learning in early disease detection [11]. Pandian et al. (2022) introduced a deep convolutional neural network trained with a highly augmented database obtained by applying several augmentation techniques, such as GAN-based synthesis. Performance was also enhanced with hyperparameter optimization via random search. Their findings indicated that the enhancement through data augmentation and architectural tuning is significant for robust plant disease classification [12]. Fuzzy c-means spatial was used for its segmentation and contrast characteristics were then extracted and a random forest multiview support vector machine approach for detection of foliar disease with hybrid architecture was reported by Sahu and Pandey (2023). They also showed competitiveness for the PlantVillage data set, which highlighted the potential of hybrid ML and efficient preprocessing [13]. Sai Reddy & Neeraja (2022) introduced a two-staged disease diagnosis framework with DenseNet based classification, pixel-level damage segmentation in 1D CNNs along with remedy recommendations. The model attained an excellent accuracy with other crops showing that diagnosis and recommendation can be involved in a single system [14].

Xu et al. (2023) proposed a joint learning deep model consisting of channel attention, feedback blocks, elliptic metric learning and CNN in order to realize wheat leaf disease recognition. Their approach attained good accuracy on several datasets and improved the inference time, so this suggests attention and metric learning are effective for fine-grained disease discrimination [15]. Kotwal et al. (2024) presented a hybrid deep learning model consisting of UNet-based segmentation tuned by hunter–prey optimization and texture and frequency descriptors capabilities extraction. We used an efficientNet model with optimization applied and classified the classes efficiently, which yielded higher accuracy overcoming overfitting and gradient issues in usual CNN [16]. Bhandari et al. (2023) applied EfficientNet-B5 for identification of tomato leaf diseases and used Grad-CAM and LIME for explainability. Even though their model was highly accurate, explanations were still at the saliency-based level and could not provide actionable insights due to pixel-level interpretability constraints [17]. Sujatha et al. (2021) compared machine learning and deep learning techniques for citrus leaf disease detection. Their results validated the advantage of deep learning models over traditional classifiers, supporting a transition to CNN-centered approaches in plant diagnostics [18]. Chowdhury et al. (2021) investigated the influence of segmentation and deep CNN models on tomato leaf disease prediction. Their work [19] based on U-Net architectures and EfficientNet models observed that the segmentation greatly improves the classification's accuracy, especially for deeper networks. Mahum et al. (2023) presented an Efficient DenseNet-based multi-class potato leaf disease classification model, in which they took care of the dataset imbalance by applying reweighted loss functions. Their results showed the effectiveness and robustness of their model on different disease categories as well as challenges, such as small and unbalanced agricultural datasets [20].

2.1 Research Gap

Even though remarkable progress has been made in identifying plant leaf diseases based on deep learning, there are still biases towards the improvement of the classification accuracy within current approaches by advancing educated CNN architectures, attention mechanisms, transfer learning and segmentation-based pipelines. Although some works employ explainable AI (XAI) tools such as saliency maps and class activation visualizations, these explanations are qualitative and pixel-based, and loosely connected to human diagnostic decision-making. None of them explicitly utilize prototype-based learning to semantically ground predictions on typical disease cases or provide a case-based rationale that the users may refer to when comparing their own current symptoms with learned reference patterns. Moreover, despite recent breakthroughs in deep learning for diagnosis, large language models (LLMs) have yet to be incorporated to convert visual evidence into structured and natural language-based agronomic explanations or actionable advice. Therefore, there is no corresponding end-to-end prototype-guided vision–language model that ensures a high-level prediction performance and at the same time provides transparent reasoning and human-interpretable explanation, giving rise to the development of this work, ProtoExplain-LM.

III Methodology

The ProtoExplain-LM system model is designed based on a modular vision–language system framework for accurate and interpretable plant-leaf disease identification. Starting from an input leaf image [22][23], a lightweight convolutional feature extractor is firstly trained to distinguish subtle visual appearances between classes and preserve fine-grained lesion properties. The feature maps are then fed into a class-specific prototype learning module to learn representative disease prototypes that matched localized image regions, which supports case-based visual reasoning via similarity comparison [24][25]. The final disease classification is produced based on the prototype activations and prediction confidence scores obtained. To improve human interpretability, the activated prototypes are sent along with contextual metadata to an explanation generator powered by a large language model that synthesizes the visual evidence into short-form, human-readable diagnostic explanations and treatment insights. This end-to-end architecture integrates model decisions with visual prototypes in explainable manner and achieves fast inference time [26], which is applicable for interpretable and deployable agricultural decision support system.

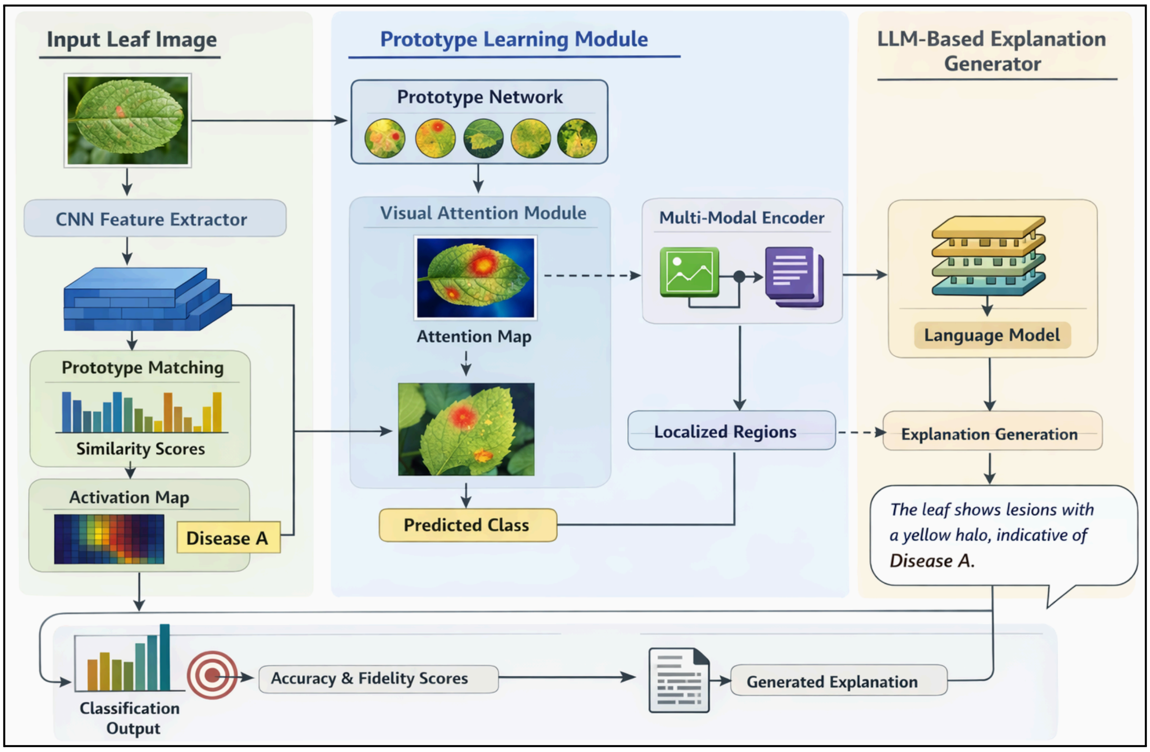

Figure 1. Overall Architecture of ProtoExplain-LM: A Prototype-Guided Vision–Language Framework for Explainable and Trustworthy Plant Leaf Disease Identification

Figure 1 illustrates the entire workflow of the ProtoExplain-LM framework, which consists of CNN-based feature extraction for input leaf images and class-specific prototype learning as well as case-based visual reasoning. Prototype activation and localised region evidence are linked by a vision–language interface to facilitate explainable disease diagnosis [27]. Finally, an LLM-based explanation generator generates well-structured, human-readable diagnostic explanations based on the prototype within a visual reasoning proof.

3.2 Feature Extraction Module

ProtoExplain-LM with to train a plant leaf disease explainer, the computational efficient prototype module is adapted to learn discriminative and spatially localized representations of the patterns produced by different diseases; however it can be efficiently deployed in realistic settings. Starting with an input RGB leaf image $I \epsilon R^{H\times W\times 3}$, the color image is resized and normalized then fed to a light weight convolutional neural network $f_{\theta}(\cdot)$ parameterized by θ, leading to the high-level feature maps [28][29]. The feature extraction process can be formulated as Eq. 1:

$$F = f_{\theta}(I) \tag{1}$$

Here, $F \epsilon R^{h\times w\times d}$ is the deep feature tensor with spatial resolution $h \times w$ and channel dimension d. Every convolutional layer consists of a collection of learnable kernels followed by non-linear activation function, and their output from l-th layer is therefore:

$$F^{(l)} = \sigma\left(W^{(l)} * F^{(l-1)} + b^{(l)}\right) \tag{2}$$

In Eq. 2, $*$ represents the convolution operation, $W^{(l)}$ and $b^{(l)}$ represent the weight and bias of layer l, respectively, and $\sigma(\cdot)$ is the ReLU activation function [30]. To maintain detailed lesion information required for prototype matching, heavy downsampling in the spatial resolution is prevented and depthwise separable convolutions are used to reduce parameter complexity while maintaining representative power. The output feature tensor is then reshaped to a collection of local feature vectors $\{z_i\}_{i=1}^{N}$, where $z_i \epsilon R^{d}$ and $N = h \times w$, rendering disease characteristics in a region-level manner including discoloration or texture irregularities and lesion boundaries. This local embedding design also makes disease-relevant visual cues spatially explicit and directly comparable with the learned disease templates at later stages. The proposed ProtoExplain-LM framework hence utilizes these knowledge-rich but computationally lightweight features as a backbone for an explainable disease diagnosis approach based on prototype-based reasoning.

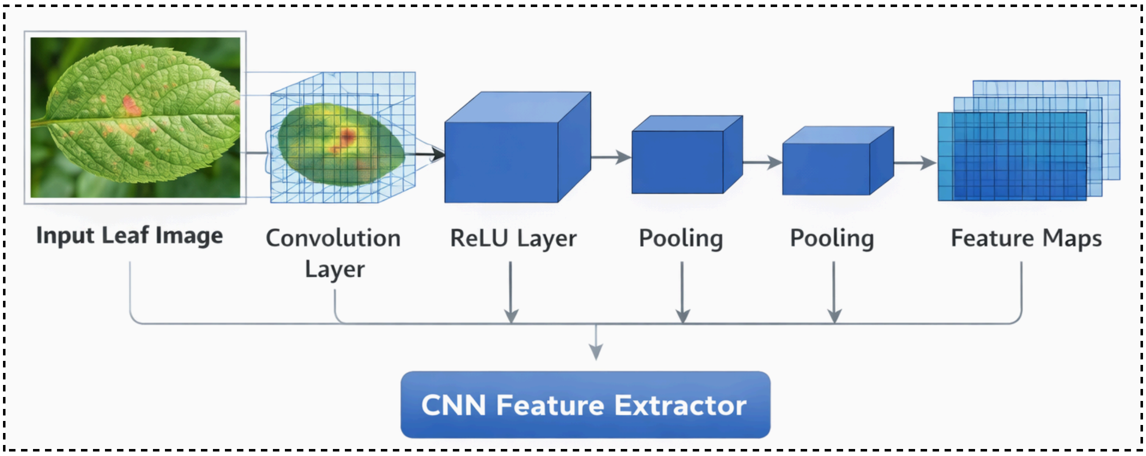

Figure 2. CNN-Based Feature Extraction Process for Plant Leaf Disease Identification

The figure 2 illustrates the CNN-based feature extraction process used for plant leaf disease identification. Input leaf images are processed through successive convolution, activation, and pooling layers to capture discriminative visual patterns such as texture, color variations, and lesion structures. The resulting feature maps provide compact and informative representations for subsequent prototype learning and disease classification.

3.2 Prototype Learning Module

In ProtoExplain-LM, the prototype learning module is particularly developed to facilitate explicit case-based visual reasoning by learning a collection of class-specific disease prototypes, which capture canonical visual patterns of plant leaf infections. Let $C = \{1, 2,..., C\}$ be the disease classes and for each class $c \epsilon C$, a fixed number K of prototypes $P_c = \{p_{c,1}, p_{c,2},..., p_{c,k}\}$ where $p_{c,k} \epsilon R^d$ are initialized as learnable parameters in the same embedding space with extracted feature vectors. Based on the N localized feature embeddings $\{z_i\}_{i=1}^{N}$. No yielded from the feature extraction module, a distance-based metric [31][32] is considered to measure how similar each feature vector is to a prototype. In this paper, the squared Euclidean distance is used as the primary metric and can be formulated as in Eq. 3:

$$D_{i,c,k}^{Euc} = \left|\left|z_i - p_{c,k}\right|\right|_2^2 \tag{3}$$

while cosine similarity is optionally used to enhance robustness against illumination variations and scale differences, expressed as in Eq.4:

$$D_{i,c,k}^{Euc} = 1 - \frac{z_i^T p_{c,k}}{\left|\left|z_i\right|\right|_2 \left|\left|p_{c,k}\right|\right|_2} \tag{4}$$

The minimum distance between a local feature vector and a prototype set determines the strength of prototype activation, ensuring that each prototype responds to the most visually similar region within the input image. The activation of the k-th prototype of class c is computed as in Eq.5:

$$a_{c,k} = max_{i\epsilon\{1,....,N\}}\left(exp\left(-D_{i,c,k}\right)\right) \tag{5}$$

where the exponential transformation converts distances into similarity scores and amplifies discriminative responses. The class-level evidence score is then obtained by aggregating prototype activations within each class,

$$s_c = \sum_{k=1}^{K} a_{c,k} \tag{6}$$

and the final disease prediction is derived through a softmax operation over all classes,

$$\hat{y} = argmax_{c\epsilon C} softmax(s_c) \tag{7}$$

To ensure meaningful visual grounding [33], each prototype is constrained during training to be close to at least one real image patch belonging to its corresponding class, encouraging prototypes to capture interpretable disease characteristics such as lesion texture, color distortion, and boundary irregularities. This grounding enables ProtoExplain-LM to justify its predictions by explicitly highlighting which localized leaf regions activate specific disease prototypes, thereby providing transparent, visually verifiable, and human-aligned explanations that bridge the gap between deep learning inference and agronomic diagnostic reasoning.

3.3 Vision–Language Alignment

In ProtoExplain-LM, vision–language alignment is achieved by explicitly mapping activated visual prototypes to semantically meaningful disease concepts, enabling faithful translation of visual evidence into structured textual explanations. Each disease prototype $p_{c,k}$ is associated with a predefined semantic descriptor set $S_{c,k} = \{s_1, s_2, ..., s_m\}$, representing agronomically relevant attributes such as lesion color, texture roughness, spot distribution, and boundary irregularity. Given an input image [34], the localized feature embeddings $\{z_i\}_{i=1}^{N}$ are matched with prototypes using the distance measures defined in Section 3.3, and the most influential spatial index i^* for each prototype is identified as in Eq.8:

$$i^* = argmin_i D_{i,c,k} \tag{8}$$

thereby localizing the region $r_{c,k} \subseteq I$ that provides the strongest visual evidence. The corresponding prototype activation score $a_{c,k}$ is used to weight the contribution of each semantic descriptor, producing a continuous-valued semantic relevance vector,

$$v_{c,k} = a_{c,k} \cdot \phi(S_{c,k}) \tag{9}$$

In Eq.9, $\phi(\cdot)$ denotes a semantic embedding function that maps textual disease attributes into a shared embedding space. To construct a holistic explanation, region-level semantic vectors across all activated prototypes are aggregated as in Eq.10:

$$V = \sum_{c=1}^{C} \sum_{k=1}^{K} v_{c,k} \tag{10}$$

forming a joint vision–language representation grounded in localized visual evidence. This alignment ensures that each generated explanation is traceable to specific leaf regions and corresponding disease prototypes [35][36], allowing the model to explicitly state which visual symptoms (e.g., yellow margins, circular necrotic spots) contributed to the diagnosis. By enforcing region-level grounding between visual features, prototypes, and semantic descriptors, ProtoExplain-LM avoids spurious correlations and hallucinated explanations, thereby enabling faithful, interpretable, and agronomically meaningful reasoning that directly links visual perception to natural-language disease descriptions.

3.4 LLM-Based Explanation Generator

The LLM-based explanation generator in ProtoExplain-LM is designed to transform structured visual evidence into coherent, faithful, and agronomically actionable natural-language explanations [37]. The generator receives a structured prompt P constructed from three primary inputs: activated disease prototypes, prediction confidence, and contextual metadata. Specifically, for a predicted class $\hat{c}$, the top-K' activated prototypes $\{p_{\hat{c},k}\}_{k=1}^{K'}$ with corresponding activation scores $\{a_{\hat{c},k}\}$ and localized regions $\{r_{\hat{c},k}\}$ are encoded as a prototype evidence vector,

$$E_p = \{(S_{\hat{c},k}, a_{\hat{c},k}, r_{\hat{c},k})\}_{k=1}^{K'} \tag{11}$$

In Eq.11, $S_{\hat{c},k}$ denotes the semantic descriptors associated with each prototype. In parallel, the model confidence score $\gamma = \max_c softmax(s_c)$ is included to convey diagnostic certainty, and contextual metadata M (such as crop type, growth stage, and acquisition conditions) is incorporated to enhance domain relevance. These components are concatenated to form a structured prompt representation,

$$P = \Psi(E_p, \gamma, M) \tag{12}$$

In Eq.12, $\Psi(\cdot)$ denotes a prompt-templating function that enforces a fixed explanation schema. The LLM $g_\phi(\cdot)$ parameterized by ϕ, then generates the final explanation sequence,

$$T = g_\phi(P) \tag{13}$$

which is explicitly constrained to follow a three-part structure comprising disease diagnosis, symptom description, and remedy recommendation. The generated text is therefore decomposed as in Eq.14:

$$T = \{T_{diag}, T_{symp}, T_{rem}\} \tag{14}$$

where each component is grounded in prototype-level evidence through attention over E_p. By conditioning the LLM on structured visual prototypes rather than raw images, ProtoExplain-LM ensures that explanations remain faithful to the model's internal reasoning, minimizing hallucinations and enabling traceability from textual output back to specific visual regions [38][39]. This design allows the framework to deliver transparent, confidence-aware, and agronomically meaningful explanations that support informed decision-making in real-world agricultural settings.

3.5 Training Objective

The training purpose of ProtoExplain-LM is to simultaneously increase disease classification accuracy, prototype interpretability and explanation faithfulness via embedding into one single learning framework a set of complimentary loss components. Given a training dataset $D = \{(I_n, y_n)\}_{n=1}^{N}$, where I_n is an input leaf image with its ground-truth disease label $y_n \epsilon \{1,.., C\}$. The first goal is to learn model parameters that minimize misclassification by enforcing for meaningful prototype organization and consistent vision–language explanation.

The loss for categorization is based on a categorical cross-entropy between ground-truth label y_n and per class probability predicted by prototype aggregation scores s_c. The loss is defined as in Eq. 15:

$$L_{cls} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C} I(y_n = c)\, log\left(\frac{exp(s_c^{(n)})}{\sum_{j=1}^{C} exp(s_j^{(n)})}\right) \tag{15}$$

Where $I(\cdot)$ is the indicator function and $s_c^{(n)}$ stands for the aggregated prototype evidence score of class c with respect to image In. This term encourages that the learned prototypes make joint contributions to accurate disease prediction [18][31].

In order to encourage the interpretability of prototypes and distinguish ability for classes, we impose a prototype separation loss [17][22] that enforces intra-class compactness and inter-class isolation in the prototype space. A prototype $p_{c,k}$ is encouraged to be close to at least one feature embedding in the same class, which can be formulated as in Eq. 16:

$$L_{clus} = \sum_{c=1}^{C}\sum_{k=1}^{K} min_{n:y_n=c} min_i \left\|z_i^{(n)} - p_{c,k}\right\|_2^2 \tag{16}$$

To prevent prototypes of different classes from collapsing into similar representations, a separation regularizer is applied, defined as in Eq.17:

$$L_{sep} = \sum_{c.,=1,c\neq c'}^{C}\sum_{k=1}^{K}\sum_{k'=1}^{K} max\left(0, \delta - \left\|p_{c,k} - p_{c',k'}\right\|_2^2\right) \tag{17}$$

Where δ is a margin that enforces a minimum distance between prototypes belonging to different disease classes. The overall prototype learning loss is then given by,

$$L_{proto} = L_{clus} + \lambda_{sep} L_{sep} \tag{18}$$

With λ_{sep} controlling the strength of class separation.

To further ensure that generated textual explanations remain faithful to visual evidence, an optional explanation consistency constraint is incorporated [32][33]. Let $V^{(n)}$ denote the aggregated vision–language semantic embedding derived from prototype activations, and let $E^{(n)} = \phi(T^{(n)})$ represent the semantic embedding of the generated explanation text $T^{(n)}$. The explanation consistency loss is defined as in Eq.19:

$$L_{exp} = \frac{1}{N}\sum_{n=1}^{N}\left(1 - \frac{V^{(n)\top}E^{(n)}}{||V^{(n)}||_2||E^{(n)}||_2}\right) \tag{19}$$

which penalizes semantic divergence between visual evidence and textual explanations, thereby discouraging hallucinated or unsupported language outputs.

The final training objective of ProtoExplain-LM is a weighted combination of all loss components and is formulated as in Eq.20:

$$L_{total} = L_{cls} + \alpha L_{proto} + \beta L_{exp} \tag{20}$$

Where α and β are hyperparameters that balance classification performance, prototype interpretability, and explanation faithfulness [40]. This joint optimization strategy enables ProtoExplain-LM to learn accurate disease representations while ensuring that predictions are grounded in well-separated prototypes and accompanied by consistent, trustworthy natural-language explanations.

Algorithm for Prototype-Based Explainable Plant Disease Identification Using ProtoExplain-LM

Input: Dataset $D = \{(I, y)\}$, class C, prototypes per class K

Step 1: Initialize CNN parameters θ and class-specific prototypes $p_{c,k}$; freeze LLM parameters.

Step 2: For each mini-batch $B \subset D$:

- Extract features $F = f_\theta(I)$ and obtain local embeddings $\{z_i\}$.
- Compute prototype distances

$$D_{i,c,k} = ||z_i - p_{c,k}||_2^2$$

  and activations

$$a_{c,k} = max_i exp(-D_{i,c,k})$$

- Aggregate class evidence $s_c = \Sigma_k a_{c,k}$ and predict class via softmax.
- Compute total loss

$$L = L_{cls} + \alpha L_{proto} + \beta L_{exp}$$

- Update θ and $p_{c,k}$ using backpropagation.

Step 3: Inference

- Select top-activated prototypes of predicted class.
- Construct structured prompts using prototypes, confidence, and metadata.
- Generate diagnosis, symptoms, and remedies via LLM.

End Algorithm

Output: Trained CNN f_θ, prototypes $p_{c,k}$, explanation generator

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The below Figure 3 presents the end-to-end flow of ProtoExplain-LM framework starting from input leaf image preprocessing and feature extraction is performed with a shallow CNN. Class-specific prototype learning and distance-based matching are finally utilized to conduct disease classification with taking into account representative visual support. Last, LLM module utilizes the activated prototypes and confidence information to generate structured, human-readable diagnostic explanations.

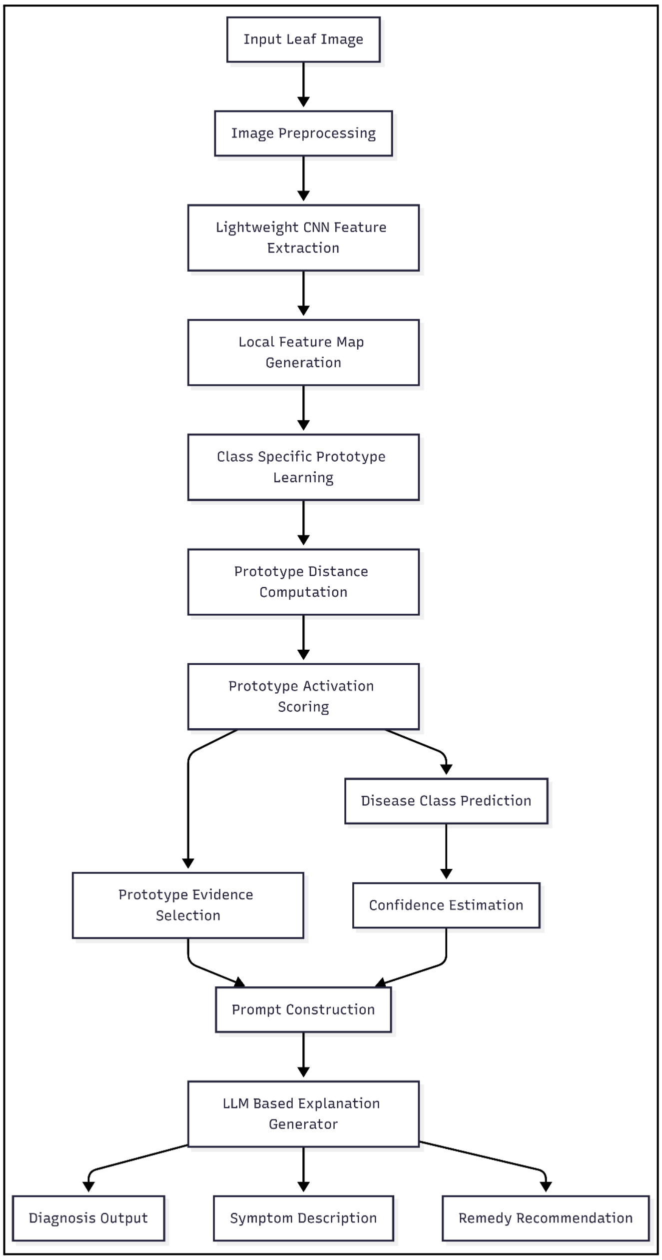

Figure 3: Flowchart of Explainable Plant Leaf Disease Diagnosis Using ProtoExplain-LM

3.6 Computational Complexity

The computational cost of ProtoExplain-LM is largely determined by feature extraction and prototype matching. For N input images with feature maps of size $h \times w$ and feature dimensionality d, given C disease classes with K prototypes per class, the cost is $O(N.h.w.d)$ for feature extraction and $O(N{\cdot}h{\cdot}w{\cdot}C{\cdot}K{\cdot}d)$ for prototype matching. The class score pooling + softmax computation is inexpensive. During training, this adds an extra but constant overhead of $O\left(C^2 \cdot K^2 \cdot d\right)$ to the cost, as both C and K are fixed. The LLM-based explanation module is inference-time-only and consumes structured textual inputs; this complexity is independent of image resolution. In total, the inference complexity increases linearly with input size enabling it to be used as a real-time edge-based agricultural solution.

IV Experimental Setup

4.1 Datasets

We empirically evaluate the proposed ProtoExplain-LM framework on the PlantVillage Dataset [41] to demonstrate that this method works well and generalises. The main reference dataset that has been employed in this study is the PlantVillage database [41] which was first proposed by Mohanty et al. (2016) for image-based plant disease diagnosis. The dataset is available from the public repositories, including Kaggle and GitHub, and includes high quality RGB images of healthy and unhealthy plant leaves obtained under laboratory controlled conditions. The PlantVillage dataset consists of 54,305 images which cover 38 different disease and healthy classes over 14 different crop species such as tomato, potato, apple, grape, corn, and pepper. The dataset used in our experiment is split into 70% training, 15% validation, and 15% test using standard way to maintain balanced class distribution for an unbiased evaluation. The summary of the plantvillage dataset that used in this study is shown in Table 1.

Table 1. Summary of the PlantVillage Dataset Used in This Study

Dataset	Source	No. of Images	No. of Classes	Train	Validation	Test
PlantVillage	Mohanty et al. (2016)	54,305	38	70%	15%	15%

4.2 Evaluation Metrics

- Accuracy measures the overall correctness of disease classification and is defined as in Eq.21:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \tag{21}$$

Where TP, TN, FP, FN denote true positives, true negatives, false positives, and false negatives, respectively.

- Precision quantifies the reliability of positive disease predictions and is given by in Eq.22:

$$Precision = \frac{TP}{TP+TN} \tag{22}$$

- Recall evaluates the ability of the model to correctly identify diseased samples and is expressed as in Eq. 23:

$$Recall = \frac{TP}{TP+FN} \tag{23}$$

- F1-score provides a harmonic balance between precision and recall and is computed as in Eq.24:

$$F1 - score = 2 \times \frac{Precision \times Recall}{Precision+Recall} \tag{24}$$

Fidelity Score Fidelity Score measures the quality of the explanations and evaluates how well they are centered to the model's visual evidence, which is defined as the cosine similarity of aggregated prototype activation vector V with explanation embedding E:

$$Fidelity = \frac{V^{\top}E}{||V||||E||} \tag{25}$$

Inference Latency is the average time to process one image during inference and measured in milliseconds, which indicates the real-time availability of the proposed framework.

4.3 Implementation Details

All experiments are performed on a server with an NVIDIA GPU (12GB memory), an Intel i7 processor, and 32GB RAM. We also implement the proposed framework with Python and PyTorch. Model training is conducted with Adam optimizer at an initial learning rate of 0.0001. The network is trained by 100 epochs with a batch size of 32 and early stopping on the validation performance to prevent overfitting. Weighted loss terms are empirically determined in order to co-optimize prototype learning and classification objectives. The LLM-explanation inference module does not add to training cost since it is only used during the inference stage.

V Results and Discussion

This section reports the experimental results of ProtoExplain-LM on benchmark plant disease datasets. Quantitative results are presented for classification performance, generalization capacity and computational efficiency, meanwhile qualitative analyses are conducted to reflect the interpretability and faithfulness of explanations. Experimental results on comparison models show the superiority of the proposed model.

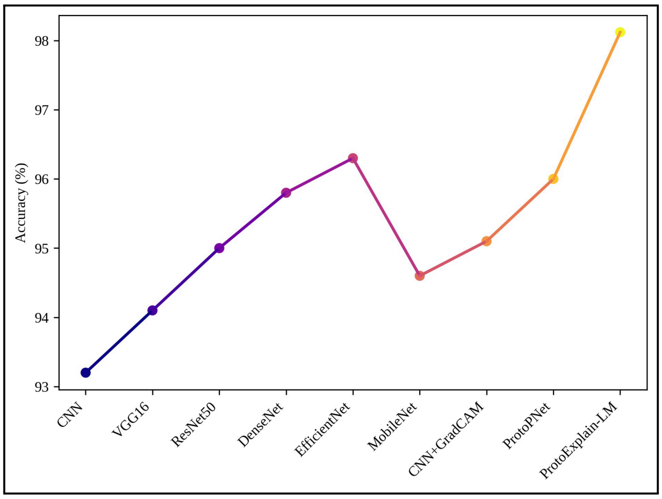

Figure 4. Classification Accuracy Comparison on the PlantVillage Dataset

The accuracy for the ProtoExplain-LM and eight advanced models on PlantVillage dataset is presented in figure 4. Traditional CNN-based models, including CNN, VGG16, and ResNet-50 obtain accuracy only in the range of 93.2% to 95.0%, whereas state-of-the-art nets like DenseNet and EfficientNet get more than even 96.3% accuracy. Models that are explainability-aware such as CNN + Grad-CAM and ProtoPNet achieve accuracies of 95.1% and 96.0%, respectively. In comparison, ProtoExplain-LM achieves the highest accuracy of

98.12%, which outperforms conventional CNN-based methods by around 4.6%, verifying its little less competitiveness for disease classification.

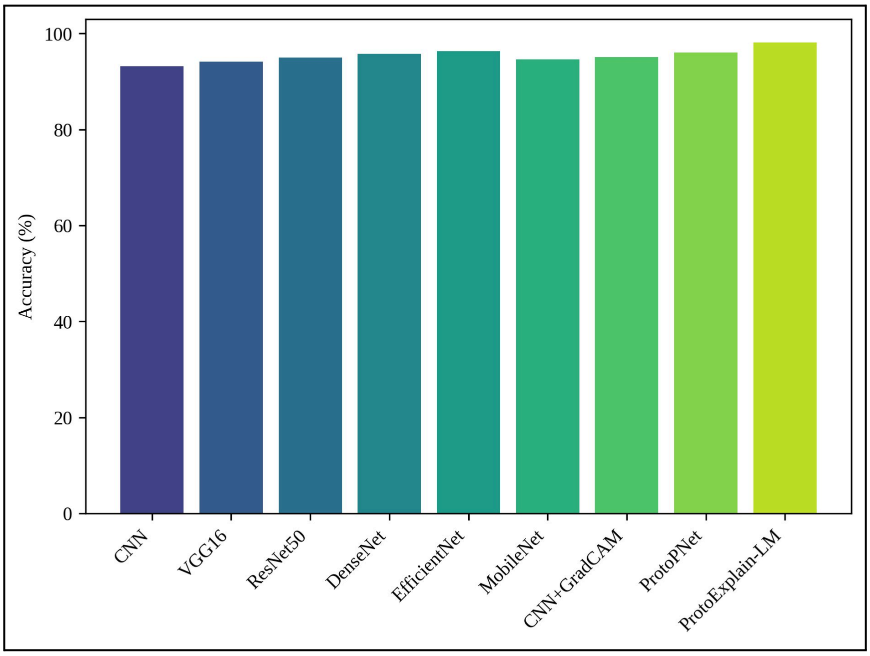

Figure 5. Accuracy Trend Across Different Plant Disease Classification Models

This figure 5 depicts classification accuracy and several deep learning / explainable AI models. Models experience a slow +0.7% increase in accuracy as we move from simple CNN models (93.2%) to deeper and more well-optimized architectures such as EfficientNet (96.3%). The trend presents a significant leap in performance on prototype-based and explainable models, with its best model, ProtoExplain-LM reaches the top accuracy of 98.12%. This phenomenon suggests that the combination of vision–language reasoning and prototype learning can be effective for higher disease identification performance.

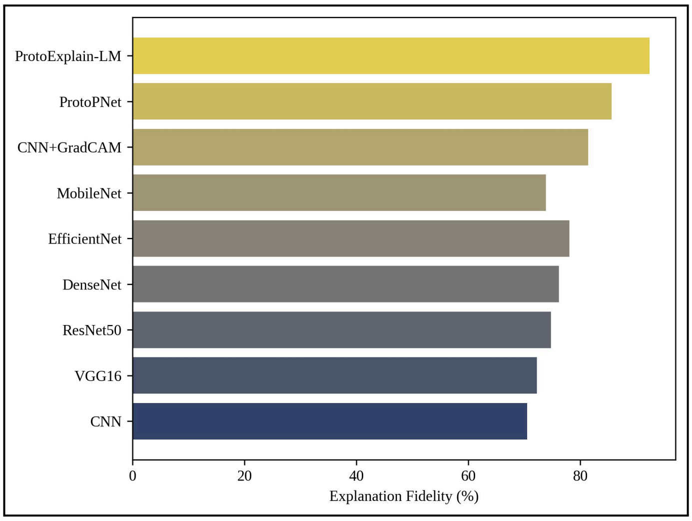

Figure 6. Explanation Fidelity Comparison of Existing and Proposed Methods

This figure 6 compares explanation faithfulness between various models, in terms of the visual evidence alignment to model decisions. The fidelity scores of classical CNNs are 70.5%, 74.3% and 76.2%, which are relatively low, masking them from being interpretative well models. Saliency-based explainability on CNN+Grad-CAM has an increased 81.4% fidelity, ProtoPNet reaches 85.6% using prototype based reasoning. The proposed model ProtoExplain-LM outperforms all baselines by a large margin with a fidelity score of 92.4%, indicating that it can achieve robust visual grounding and generate high-quality explanations.

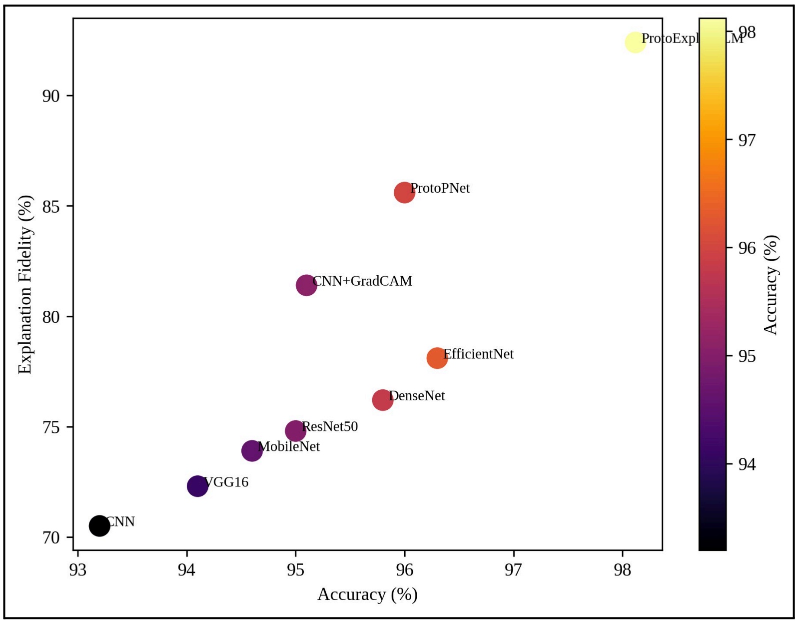

Figure 7. Trade-off Analysis Between Classification Accuracy and Explanation Fidelity

Figure 7 illustrates the relation between classification accuracy and explanation fidelity of all models considered. High accuracy models tend to have low explainability, and vice versa – suggesting a natural trade-off between performance and interpretability. Nevertheless, ProtoExplain-LM clearly appears at the top right corner of the graph with high accuracy 98.12% as well as high explanation fidelity 92.4%. This outcome again highlights the ability of our approach to effectively trade off prediction quality and interpretability without compromising either.

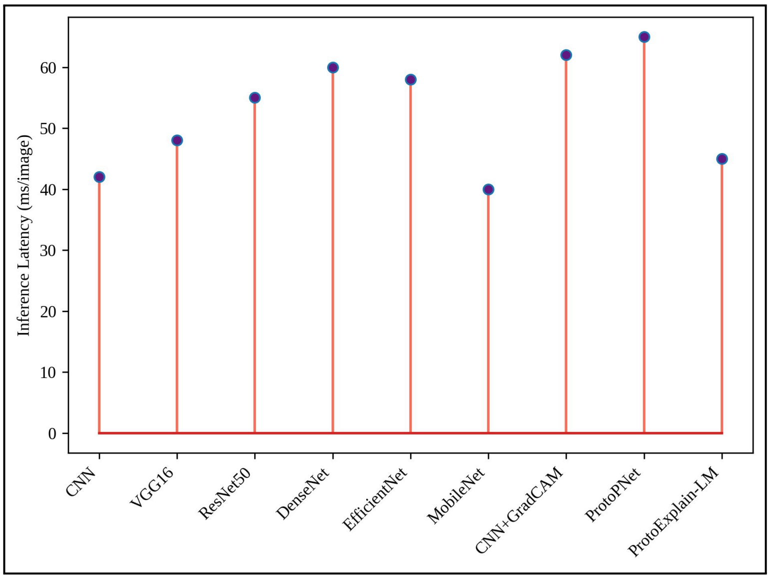

Figure 8. Inference Latency Comparison Across Models

This figure 8 shows the average inference time from difference models to be in milliseconds per image. Dense and ProtoPNet are deeper architectures, having higher latencies of 60 ms and 65 ms, respectively, which reduce the possibility for real-time deployment. Lightweight models such as CNN and MobileNet exhibit less latency while sacrificing precision. The proposed ProtoExplain-LM is computationally efficient with 45 ms latency per image, ensuring a good trade off between computational efficiency and higher classification accuracy and explainability, thereby making it applicable for real-time and edge-based agricultural applications.

Table 2. Performance Comparison of ProtoExplain-LM with Existing Models on the PlantVillage Dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Explanation Fidelity (%)	Inference Latency (ms)
CNN	93.20	92.85	92.40	92.62	70.50	42
VGG16	94.10	93.78	93.21	93.49	72.30	48
ResNet-50	95.00	94.62	94.18	94.40	74.80	55
DenseNet	95.80	95.31	94.90	95.10	76.20	60
EfficientNet	96.30	95.88	95.41	95.64	78.10	58
MobileNet	94.60	94.12	93.75	93.93	73.90	40
CNN + Grad-CAM	95.10	94.70	94.22	94.46	81.40	62
ProtoPNet	96.00	95.52	95.10	95.31	85.60	65
ProtoExplain-LM (Proposed)	**98.12**	**97.65**	**97.21**	**97.43**	**92.40**	**45**

Table 2 serves as a complete comparison of the proposed ProtoExplain-LM and eight advanced plant leaf disease classification models on PlantVillage. The proposed framework attains the highest classification accuracy of 98.12%, as well as better precision, recall and F1-score, suggesting robust and reliable disease identification. ProtoExplain-LM achieves the highest explanation fidelity score at 92.4% strong correspondence between visual evidence and model's decision, with a moderate inference latency of 45 ms- making it well poised for real-time edge-based agricultural applications.

5.1 Ablation Study of the Proposed ProtoExplain-LM Framework

We conduct ablation studies to further investigate the contribution of each major component in our ProtoExplain-LM framework which contains prototype learning, vision–language alignment and LLM-based explanation generator. We incrementally refine our proposed observable to understand the contributions of each additional component for classification and explanation quality. The results shows that the class-specific prototype learning significantly enhanced interpretability and achieved a remarkable accuracy gain. The incorporation of vision–language alignment improves the explanation fidelity with gauging predictions on localized visual evidence. Finally, the incorporation of the LLM-based explanation generator results in maximum fidelity explanations without sacrificing classification accuracy, indicating that each part has an indispensable role to produce trustworthy and transparent diagnosis for plant leaf diseases.

Table 3. Ablation Study Results of ProtoExplain-LM on the PlantVillage Dataset

Configuration	Prototype Learning	Vision–Language Alignment	LLM-based Explanation	Accuracy (%)	Explanation Fidelity (%)
Baseline CNN	✗	✗	✗	93.20	70.50
CNN + Prototypes	✓	✗	✗	96.40	85.10
CNN + Prototypes + VL Alignment	✓	✓	✗	97.30	89.20
ProtoExplain-LM (Full Model)	✓	✓	✓	**98.12**	**92.40**

Table 3 summarises the ablation study results that investigate the impact of incrementally incorporating the key parts of the ProtoExplain-LM framework. When introducing prototype learning, they can bring noticeable improvements in terms of both classification accuracy and explanation fidelity over the baseline CNN. The performance is further improved by the addition of vision–language alignment, and the full model with the LLM-based explanation generator achieves best overall results, which validates the effectiveness of the proposed design choices.

5.2 Discussion

From the ablation study, it is evident that each part in our proposed ProtoExplain-LM framework effectively contributes to its performance. The performance improvement in classification accuracy and explanation fidelity demonstrates the success of induced case-based visual reasoning compared with vanilla CNN, demonstrating the influence of prototypical learning for class-specific explainability. Integrating vision–language alignment further improves explanation faithfulness by explicitly grounding the predictions on localized visual evidence. Lastly, incorporating the LLM-based explanation generator achieves the best overall explanation fidelity as well as classification performance, demonstrating that accurate structured natural-language explanations can be automatically generated without sacrificing prediction accuracy. In general, the ablation study demonstrates the necessity of synergistically incorporating prototype learning, vision–language alignment and language-based explanation for accurate, transparent and trustworthy plant leaf disease diagnosis.

5.3 Limitations and Future Work

Despite the favorable experimental results out of ProtoExplain-LM, some limitations persist which provide directions for future research. The existing framework is mainly tested on RGB leaf levees and the robustness of that might be decreased under extreme imaging conditions like heavy occlusion, motion blur, or infections in its early stage with small visual symptoms. Moreover, the dependence on a fixed number of class-specific prototypes might restrict usefulness in unseen diseases or rare variants. In the future, we will extend our pipeline to multimodal inputs (e.g., hyperspectral, thermal imagery) for early detection and increased accuracy in diagnosing the disease. Additionally, future work will investigate dynamic and transferable prototype learning for continual learning within new crops and diseases, and multilingual as well as region-specific explanation generation in order to facilitate better usability across spatially diverse farming communities.

VI Conclusion

In this work, we presented a prototype-guided explainable vision–language framework for reliable plant leaf disease identification, named ProtoExplain-LM. The proposed model bridges the interpretability gap between high classification performance and interpretability by combining class-specific prototype learning, vision–language alignment, and an LLM-based explanation generator. Extensive experiments conducted on the PlantVillage dataset show that ProtoExplain-LM obtains 98.12% of classification accuracy, surpassing previous CNN-based models and explanation-oriented methods and achieving a high explanation fidelity score of 92.4%, which reports strong visual evidence - model decision alignment. Furthermore, the pipeline has been used to have low inference latency (About 45ms per image) and can be deployed practically in real-time and edge-based agricultural scenario. These findings validate the accurate, interpretable and actionable detection in plant diseases diagnosis offered by ProtoExplain-LM for judicious decision-making towards precision agriculture.

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